INCH-POUND

MIL-C-27072/77B
10 May 1994
SUPERSEDING
MIL-C-27072/77A
24 September 1987

MILITARY SPECIFICATION

CABLE, SPECIAL PURPOSE, ELECTRICAL, MULTICONDUCTOR, (2 SHIELDED CABLES AND 19 WIRES UNDER A DOUBLE SHIELD) M27072/77MACA*

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and the latest issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-27072.

(B)

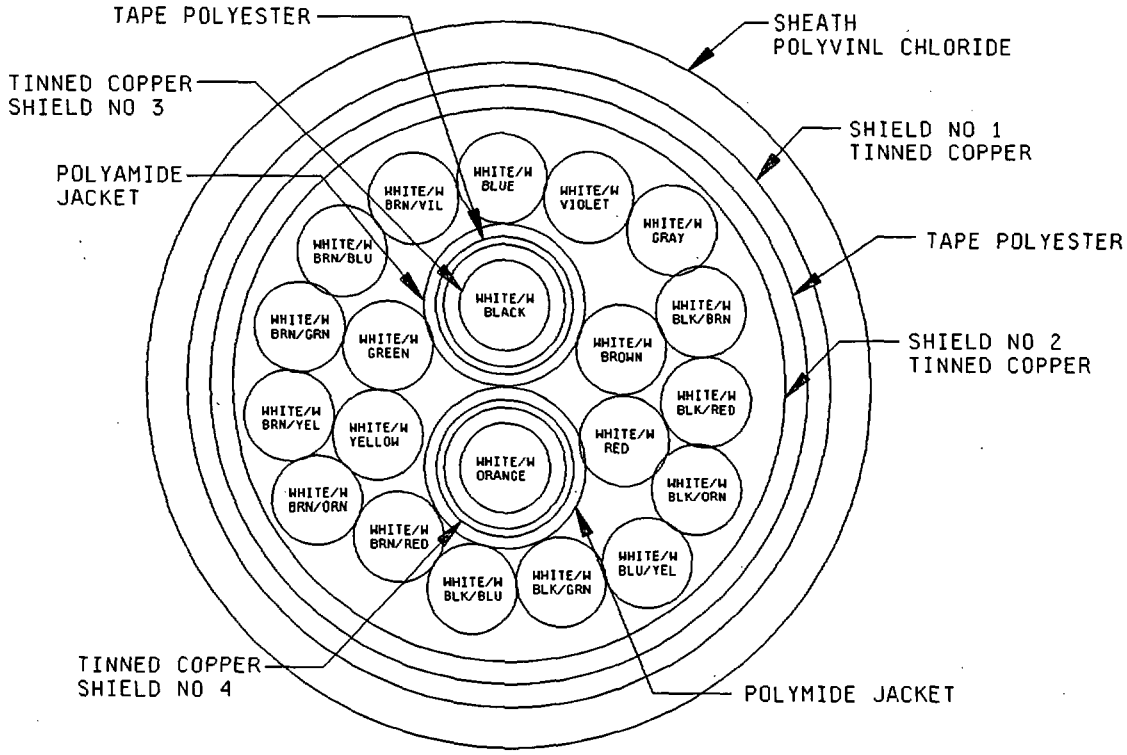


FIGURE 1. Cable configuration.

* The complete PIN requires a color and code designator (see table I).

(B) denotes changes

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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DESIGN DATA:

Design and construction: See figure 1.

Number of conductors: Twenty-one

Conductor size and stranding: See table I.

Wire: In accordance with M16878/17BFB*.

Jacket undershield: Not applicable.

Shielding: Tinned copper.

Sheath: Polyvinyl chloride.

Temperature rating: 105°C.

REQUIREMENTS:

Visual and dimensional:

Components:

Basic wire: Tinned copper.

Dimension: See table I.

Primary insulation:

Material: Polyvinyl chloride.

Wall thickness: .010 inch ±.003 inch.

Colors:

White with black tracer.

White with brown tracer.

White with red tracer.

White with orange tracer.

White with yellow tracer.

White with green tracer.

White with blue tracer.

White with violet tracer.

White with gray tracer.

White with black and brown tracers.

White with black and red tracers.

White with black and orange tracers.

White with black and yellow tracers.

White with black and green tracers.

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White with black and blue tracers.

White with brown and red tracers.

White with brown and orange tracers.

White with brown and yellow tracers.

White with brown and green tracers.

White with brown and blue tracers.

White with brown and violet tracers.

Spark test voltage: 3,400 V rms.

Inspection test voltage: 2,000 V rms.

Impulse dielectric test: 8,000 V peak.

Jacket over insulation: Polyamide.

Wall thickness: .003 inch ±.001.

Color: Natural.

Filler: Not applicable.

Tape material: Polyester.

Cabling:

Core:

Number of conductors: Six.

Shielded conductors:

Number of conductors: Two.

Shield: (Shield numbers 3 and 4).

Material: Tinned copper.

Gauge: 34 AWG.

Braid angle: 20 to 40 degrees.

Tape over shield: Polyester material in accordance with MIL-C-27072.

Tape thickness: .002 inch minimum.

Tape overlap: 25 percent.

Jacket over tape: Polyamide in accordance with MIL-C-27072.

Unshielded conductors:

Number of conductors: Four.

Pitch or lay: 2.0 inches nominal.

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First layer:

Number of conductors: Fifteen.

Pitch or lay: 6.0 inches nominal.

Fillers: Optional.

Shield: Number 2.

Material: Tinned copper.

Gauge: 34 AWG.

Braid angle: 20 to 40 degrees.

Tape: Polyethylene terephthalate.

Shield: Number 1.

Material: Tinned copper.

Gauge: 34 AWG.

Braid angle: 20 to 40 degrees.

Workmanship: In accordance with MIL-C-27072.

Sheath:

Material: Polyvinyl chloride.

Wall thickness: .040 inch.

Tensile strength: In accordance with table V of MIL-C-27072.

Elongation: In accordance with table V MIL-C-27072.

Tension set: Not applicable.

Tear strength: Not applicable.

Accelerated aging: In accordance with table V of MIL-C-27072.

Oil resistance: Not applicable.

Color: See table IV of MIL-C-27072.

Finished cable:

Diameter: In accordance with table I.

Conductor resistance: In accordance with MIL-W-16878/17.

Dielectric strength: In accordance with MIL-W-16878/17.

Cold bend: Mandrel size in accordance with table XIII of MIL-C-27072.

Ozone: Mandrel size in accordance with table XIV of MIL-C-27072.

Marking durability: In accordance with MIL-C-27072.

Heat stability: Not applicable.

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TABLE I. Construction data.

PIN M27072/77	Conductor				Finished cable diameter (in.) ±10%	Basic wire M16878/17
	Wire size (AWG)	Number of strands	Strand size (AWG)	Nominal dia. (in.)		
MACA*	22	7	30	.056	.50	BFB***

Asterisks (*) shall be replaced with the appropriate color designator.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85

Review activities:
Air Force - 85
DLA - ES, IS

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6145-2067-09)

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